University Of Engineering and Technology, Lahore
Computer Engineering Department

Lab Evaluation:

Course Name: Data Structures & Algorithm	**Course Code:** CMPE-212L
Assignment Type: Semester Project	**Dated:** 30th April 2024
Semester: 4th	**Session:** 2022
Lab/Project/Assignment #: 1	**CLOs to be covered:** CLO 2
Lab Title: Spell Corrector	**Teacher Name:** Engr. Afeef Obaid

CLO 2	Design solutions for problems using appropriate data structures and algorithms.		
Levels (Marks)	**Level 1 Good**	**Level 2 Average**	**Level 3 Bad**
Criteria 1			
Criteria 2			
Criteria 3			
Criteria 4			
		Total	**/40**

Rubrics for Current Lab Evaluation

Criteria	Good (10-8)	Average (7-4)	Bad (3-1)
Appropriate data structure is selected for dictionary	BST or Hash-table is used	array is used	directly matched from file
Spell checker	spell checker works on stem words	spell checker identifies correct and incorrect spellings and writes to file	spell checker identifies correct and incorrect spellings
Spell corrector	shows options and store to file selected option	don't show auto-generated options but stores the user told to file / just shows auto generated options.	don't store user told options to file neither auto-generate them
Completeness	Complete correct file is generated	only spell checker is implemented	only dictionary is loaded

Semester Project

Spell Corrector

Spell correctors can work in various ways. In our version of a spell corrector, the program will read a text file to be checked and produce another file with the correctly spelled text in it. Any given word in file can either be correctly spelled or incorrectly spelled.

To do:

1. **Dictionary:** Program should read the provided dictionary dict.txt from file, and insert it into appropriate data structure.

2. **Spell Checker:** After loading the dictionary, spellChecker will read the file input and each of its words will be checked with help of dictionary if the word is found it considered correctly spelled, and written in correct file.

3. **Spell Corrector:** What is a spell checker if it can't give you the alternate options for wrong spellings. For spell corrector, after finding words that arn't correct, find the correct alternates and ask the user which one they prefer and write that in correct file.

4. To increase the hit rate of spellCheck it's advisable that words should be reduced to their roots as most dictionaries only have root words.This will require some preprocessing on the word to be checked according to the language rules governing this process.

   Preprocessing

   Convert the word into lowercase
   - If the word ends in -ing, remove the -ing
   - If the word ends in -s, remove the -s
   - If the word ends in -es, -ly, or -ed, remove the -es, -ly, or -ed
   - If the word ends in -ing, remove the -ing and add -e
   - If the word ends in -ies, remove the -ies, and add -y
   - If the word ends in -es or -ed, remove the –s or –d